

Detecting Directional Coupling in Network Dynamical Systems via Kalman's Observability

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Detecting coupling in network dynamical systems from time series is an open problem in the physics of complex systems. In this Letter, we tackle this issue from a control-theoretic perspective. Drawing inspiration from Kalman's notion of observability, we argue the presence of directional coupling between two units, $X \rightarrow Y$, when X is detected as an internal state from the measurement of Y . We illustrate this approach on a series of analytically tractable systems, showcasing how it overcomes limitations of state-of-the-art methods for network inference.

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Introduction—Emergent phenomena in complex systems offer a glimpse into how interactions between simple units can give rise to intricate, large-scale behaviors [1–3]. These patterns are ubiquitous in physics, from the synchronization of coupled oscillators [4,5] to the magnetization of ferromagnetic materials [6]. Unraveling the pairwise interactions that underpin collective behavior from observations remains an open challenge [7]. Determining whether two units in a collective are coupled based on their dynamics is a question that continues to intrigue both physicists and mathematicians. This problem has been studied exhaustively [8–13], but we still lack tools that can solve even rudimentary problems. For example, “given the correlations between just two classical variables, it is impossible to determine whether they arose from a causal influence of one on the other or a common cause influencing both” (that is, causal sufficiency) [14].

Information theory is the gold standard for the inference of coupling between units from their time series. The most famous measure is transfer entropy [15,16], which measures statistical dependence between dynamical units, akin to Granger causality [17]. Over the years, transfer entropy has seen many technical improvements, such as escaping the curse of dimensionality [18]. It has also been integrated with the work on causality pioneered by Pearl [19], leading to the development of the PCMC algorithm and its descendants [20,21]. While these model-free algorithms

are generally effective, they rely on assumptions, such as faithfulness or causal sufficiency, which are strained in many real-world applications.

In this Letter, we introduce a new method for detecting coupling between units from time series. Our approach is inspired by recent advancements at the intersection of physics, network science, and control theory, particularly by the concept of Kalman's observability [22–25] and its use in network dynamical systems [26–28]. Control theorists use sensors to measure some of the system's external states and then, along with a mathematical model, estimate internal states to apply control inputs. In our method, we argue that the ability to observe internal states from external ones implies a coupling between internal and external states, so that the notion of observability is leveraged for structure inference. We demonstrate this approach for analytically tractable models of collective dynamics, overcoming pitfalls of the state of the art.

Theory—We represent collective dynamics as a network dynamical system with nodes representing individual states and directed edges representing coupling between them [29]. Specifically, we consider an N node, network dynamical system of the form

$$\dot{\mathbf{X}}_t = \mathcal{A}\mathbf{X}_t, \quad \mathbf{Y}_t = \mathcal{C}\mathbf{X}_t, \quad (1)$$

where $\mathbf{X}_t \in \mathbb{R}^{N \times 1}$ is the complete state vector (comprising internal and external states) at time t , $\dot{\mathbf{X}}_t$ is its time derivative, and $\mathbf{Y}_t \in \mathbb{R}^{M \times 1}$ is the external state vector.

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$\mathcal{A} \in \mathbb{R}^{N \times N}$ is the state transition matrix encapsulating the network topology, such that $\mathcal{A}[ij]$ is the weight of the edge from node j to node i , and $\mathcal{C} \in \mathbb{R}^{M \times N}$ is the output matrix that maps some of the states to the external ones.

The notion of observability is used to describe the extent to which the internal states of the system can be estimated from the external ones. One way to study observability is through the observability matrix, $\mathcal{O} = [\mathcal{C}^T, (\mathcal{C}\mathcal{A})^T, \dots, (\mathcal{C}\mathcal{A}^{N-1})^T]^T$, whose rank [denoted by $\rho(\cdot)$] is equal to the dimension of the orthogonal complement to the unobservable subspace. Obviously, observability depends on which states are external, a fact we leverage to design the proposed method. Let $\mathcal{C}_i$ be the output matrix corresponding to node i being the only external state and $\mathcal{C}_{ij}$ be the output matrix corresponding to nodes i and j being the only external states; the rows of $\mathcal{C}_i$ and $\mathcal{C}_{ij}$ are standard basis vectors. Let also $\mathcal{O}_i$ and $\mathcal{O}_{ij}$ be the observability matrices corresponding to $\mathcal{C}_i$ and $\mathcal{C}_{ij}$, respectively. Assuming that $\mathcal{A}$ has a nonzero diagonal, we can formulate the following theorem, whose proof is in the Supplemental Material [30].

Theorem 1—There is a directional coupling from i to j ($i \rightarrow j$) if and only if $\rho(\mathcal{O}_j) = \rho(\mathcal{O}_{ij})$ with probability one.

Given Theorem 1, we propose to infer coupling (or lack of) between units from the dimension of the corresponding observable spaces, as illustrated in the following case.

Illustration of the theory—Consider a network dynamical system with three units, Z, X, and Y ($\mathbf{X}_t = [Z_t, X_t, Y_t]^T$), with state matrix

$$\mathcal{A} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \quad (2)$$

such that true structure is $X \leftarrow Z \rightarrow Y$; here and in what follows, we use roman font for the units in the collective. If we wish to detect the existence (or lack of) coupling between Y and Z, for example, we inspect the rank of the three observability matrices $\mathcal{O}_Z$, $\mathcal{O}_Y$, and $\mathcal{O}_{ZY}$. With simple algebra, we obtain $\rho(\mathcal{O}_Z) = 1$ for $\mathcal{C}_Z = [1 \ 0 \ 0]$, $\rho(\mathcal{O}_Y) = 2$ for $\mathcal{C}_Y = [0 \ 0 \ 1]$, and $\rho(\mathcal{O}_{ZY}) = 2$ for $\mathcal{C}_{ZY} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. According to Theorem 1, we determine $Z \rightarrow Y$ only by comparing the rank of the three observability matrices. A similar argument can be carried out for any other pair of units, ultimately unveiling the exact network topology.

Detection matrix—In real-world scenarios, we have no information about $\mathcal{A}$, but we only have access to time series of the external states $\mathbf{Y}$, so that Theorem 1 cannot be directly applied. To overcome the issue, we utilize the so-called detection matrix, $\mathcal{T}_\Omega \in \mathbb{R}^{MK \times L}$, proposed by Haehne *et al.* [26] to identify the number of hidden nodes in a network dynamical system. The detection matrix is built by stacking K measurements (sampled at a constant rate) from the set of measured nodes Ω of cardinality M , for L different initial conditions,

$$\mathcal{T}_\Omega = \begin{bmatrix} \mathbf{Y}_1^{(1)} & \mathbf{Y}_1^{(2)} & \dots & \mathbf{Y}_1^{(L)} \\ \mathbf{Y}_2^{(1)} & \mathbf{Y}_2^{(2)} & \dots & \mathbf{Y}_2^{(L)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{Y}_K^{(1)} & \mathbf{Y}_K^{(2)} & \dots & \mathbf{Y}_K^{(L)} \end{bmatrix}. \quad (3)$$

Porfiri [27] uncovered a link between the detection matrix and the observability matrix, namely,

$$\rho(\mathcal{T}_\Omega) = \rho(\mathcal{O}_\Omega). \quad (4)$$

This equality offers a valuable point for inferring the structure of the network when combined with Theorem 1. The proposed method of detecting the directional coupling in network dynamical systems requires only building and computing the rank of detection matrix for the units in question. We note that directional coupling may be indirect, whereby the statement in Theorem 1 does not specify the length of the path between nodes i and j .

Illustration of the proposed method—Revisiting the example above, we construct three detection matrices from three system realizations ($L = 3$), each based on four data points ($K = 4$) sampled one time unit apart, with basis vectors as initial conditions. In agreement with analytical results, we obtain $\rho(\mathcal{T}_Z) = 1$, $\rho(\mathcal{T}_Y) = 2$, and $\rho(\mathcal{T}_{ZY}) = 2$, and conclude directional coupling from Z to Y from data only (Fig. 1). In the Supplemental Material [30], we present a numerical study on larger networks [31,32].

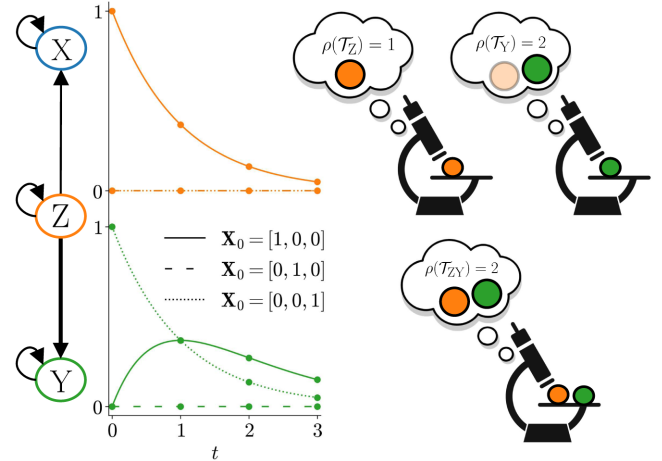


FIG. 1. Illustration of the approach to detect the coupling between Z and Y in system (2), by sampling four data points from the time series of Z and Y across three distinct initial conditions, denoted as $\mathbf{X}_0$. We construct the detection matrices $\mathcal{T}_Z$, $\mathcal{T}_Y$, and $\mathcal{T}_{ZY}$: from the rank of these matrices, we determine directional coupling, $Z \rightarrow Y$. The microscopes on the right provide some intuition about the method, whereby we uncover an internal state when measuring Y, but not when measuring Z or Y and Z together.

Application of the method to the Ornstein-Uhlenbeck system—The detection matrix works perfectly for linear, noise-free dynamical systems [27], yet, it can be applied with reliable outcomes to more complex cases [26–28]. Similar to Runge *et al.* [18], we consider the classical Ornstein-Uhlenbeck stochastic dynamical system,

$$d\mathbf{X}_t = \mathcal{A}\mathbf{X}_t dt + d\mathbf{W}_t, \quad (5)$$

where $\mathbf{W}_t$ is a multivariate Wiener process with identity covariance matrix representing uncorrelated Gaussian noise. The Ornstein-Uhlenbeck system was introduced by Langevin in his seminal work on Brownian motion [33]. Just as Brownian motion is the scaling limit of a random walk, the Ornstein-Uhlenbeck system is the limit of the Ehrenfest urn model for the diffusion of gas particles [34].

While the detection matrix is not directly applicable to this case, the mean of the system evolves in time according to a linear network dynamical system with state matrix $\mathcal{A}$ [35] (see Supplemental Material [30]). Thus, building the detection matrix using ensemble averages of R realizations of the stochastic system is equivalent (for large R) to constructing the matrix from the noiseless system. We can infer the structure of the system from the detection matrix, whereby numerical results [Fig. 2(a)] yield $\rho(\mathcal{T}_Z) = 1$, $\rho(\mathcal{T}_X) = 2$, $\rho(\mathcal{T}_Y) = 2$, $\rho(\mathcal{T}_{ZX}) = 2$, $\rho(\mathcal{T}_{ZY}) = 2$, $\rho(\mathcal{T}_{XY}) = 3$, and $\rho(\mathcal{T}_{ZXY}) = 3$ (see Supplemental Material [30]). These results show that the proposed method can correctly infer the structure of this simple triad, but so do standard methods, such as partial correlation (see Supplemental Material [30]).

Instead, for the scenario where Z is hidden (that is, causal sufficiency is in question), standard techniques would fail to identify the lack of coupling between X and Y (see Supplemental Material [30]). It can be analytically shown [36] that transfer entropy from X to Y is $\text{TE}_{X \rightarrow Y} = 0.0054$ nats and mutual information between X and Y $\text{MI}_{X,Y} = 0.06$ nats. Numerically, the LPCMCI algorithm, the state-of-the-art algorithm in identifying causal structures in the presence of hidden units [21], fails to infer the lack of coupling between X and Y . Using the proposed framework, we conclude that X and Y are decoupled since $\rho(\mathcal{T}_X) = \rho(\mathcal{T}_Y) = 2 \neq \rho(\mathcal{T}_{XY}) = 3$.

Intuition behind the method—Some intuition about the proposed method is built by stressing the distinction between internal and external states. With reference to the example in Fig. 1, the fact that $\rho(\mathcal{T}_Y) = 2$ indicates that the time evolution of Y depends on one other, internal, state. This is not the case for Z , whereby we find $\rho(\mathcal{T}_Z) = 1$ so that the time evolution of Z unfolds independently of other nodes in the system. Measuring Y and Z together, we determine $\rho(\mathcal{T}_{ZY}) = 2$, suggesting the absence of other internal states underpinning the time evolution of these two states. Based on these observations, we conclude that Z is directionally coupled to Y . Similar arguments can

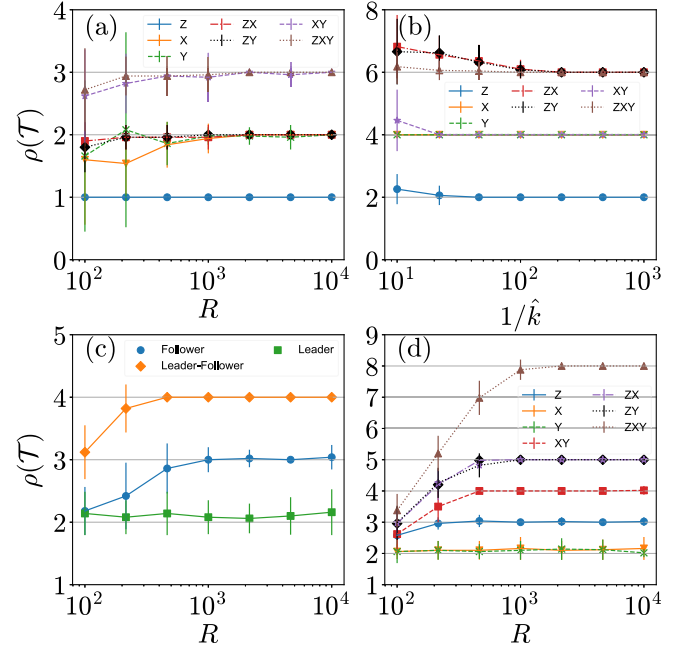


FIG. 2. Rank of the detection matrix as a function of, the number of realizations R for (a),(c),(d) and the magnitude of nonlinearity for (b). Error bars represent the standard deviation over 50 trials. (a) Results for the Ornstein-Uhlenbeck system ($L = 9$, $K = 15$). (b) Results for Duffing oscillators ($L = 9$, $K = 15$, sampling rate is one data point per unit time). (c) Results for the leader-follower model with $L = 15$, $K = 30$, $\eta_F = 0.2$, $\eta_L = 0.8$, and $w = 0.5$. (d) Results for the synergistic system, $L = 15$, $K = 30$ when measuring a single unit, $L = 20$, $K = 40$ when measuring two units, and $L = 25$, $K = 45$ when measuring all three units. In (a) and (b), we utilized the largest gap between the logarithm of the singular values of matrices as a threshold to determine their rank, while in (c) and (d), we used the method proposed by Succar *et al.* [28] to compute the rank.

be carried out for more complex dynamics, such as non-autonomous systems and synchronous dynamics [37,38] (see Supplemental Material [30]).

Extension to weakly nonlinear oscillators—We consider three nonlinear mechanical oscillators: two coupled oscillators, X and Y , and an independent oscillator, Z , with their dynamics governed by the Duffing equation [39],

$$\ddot{z} + \dot{z} + \hat{k}z^3 + z = 0, \quad (6a)$$

$$\ddot{x} + 2\dot{x} - \dot{y} + \hat{k}(x^3 - (y-x)^3) + 2x - y = 0, \quad (6b)$$

$$\ddot{y} + 2\dot{y} - \dot{x} + \hat{k}(y^3 + (y-x)^3) + 2y - x = 0. \quad (6c)$$

The parameter $\hat{k}$ quantifies the extent of nonlinearity. This system allows for testing the validity of the method in weakly nonlinear multidimensional systems. We build the detection matrix from the time evolution of variables z and $\dot{z}$ for Z , x and $\dot{x}$ for X , and y and $\dot{y}$ for Y , so that each state is two dimensional. For weak nonlinearity,

we obtain $\rho(\mathcal{T}_Z) = 2$, $\rho(\mathcal{T}_X) = \rho(\mathcal{T}_Y) = \rho(\mathcal{T}_{XY}) = 4$, and $\rho(\mathcal{T}_{ZX}) = \rho(\mathcal{T}_{ZY}) = \rho(\mathcal{T}_{XYZ}) = 6$, so that we correctly infer the coupling between X and Y and their independence of Z [Fig. 2(b)].

Extension to Markov chains—Markov chains are instrumental in physics for modeling phenomena like phase transitions or atomic emissions, where systems undergo sequential changes between distinct states [40]. While the basic incarnation of the detection matrix [26] does not apply to Markov chains, we have recently provided an extension to hidden Markov chains [28] that can be adapted to support the implementation of the proposed method, as detailed below.

We consider a discrete-time Markov chain, $\mathbf{S}$ with internal states taking values in a discrete space $\{s_1, \dots, s_N\}$ with probability mass function (pmf) $\mathbf{X}_k = [\Pr(S_k = s_1), \dots, \Pr(S_k = s_N)]^T$. The time evolution of the pmf is

$$\mathbf{X}_{k+1} = \mathcal{P}^T \mathbf{X}_k, \quad (7)$$

where $\mathcal{P} \in [0, 1]^{N \times N}$ is the row-stochastic transition matrix. A corresponding hidden Markov chain $\mathbf{H}$, would be an output representation of $\mathbf{S}$ where some of the internal states are indistinguishable. Therefore, $\mathbf{H}$ takes values in $\{h_1, \dots, h_M\}$, such that

$$\mathbf{Y}_k = \mathcal{C} \mathbf{X}_k, \quad (8)$$

where $\mathbf{Y}_k = [\Pr(H_k = h_1), \dots, \Pr(H_k = h_M)]^T$ is the pmf of the hidden Markov chain. The system of equations in Eqs. (7) and (8) resemble Eq. (1), such that building the detection matrix from the pmf hidden Markov chain ($\mathbf{Y}_k$) and computing its rank would yield the dimension of the observable subspace of $\mathbf{S}$.

Application of the method to Markovian leader-follower dynamics—We introduce a modification of the random Boolean model of collective behavior proposed by De Lellis *et al.* [41],

$$L_{k+1} = \begin{cases} L_k, & \text{w.p. } 1 - \eta_L \\ 1 - L_k, & \text{w.p. } \eta_L, \end{cases} \quad \text{and} \quad (9a)$$

$$F_{k+1} = \begin{cases} F_k, & \text{w.p. } |1 - L_{k+1} - F_k|w + (1 - \eta_F)(1 - w) \\ 1 - F_k, & \text{w.p. } |L_{k+1} - F_k|w + \eta_F(1 - w), \end{cases} \quad (9b)$$

where w.p. stands for “with probability.” In this model, the leader L switches between 0 and 1 at each time step with probability $0 < \eta_L < 1$, while the follower F switches states with probability $0 < \eta_F < 1$ in the absence of the gain $0 \leq w \leq 1$, which is the tendency of the follower to follow the leader.

The instantaneous coupling between the units challenges the use of methods based on sequential cause and effect such as transfer entropy. In our framework, we can easily

infer the direction by simply constructing the detection matrices (see Supplemental Material [30]). For a sufficiently large number of realizations of the Markov chain, we obtain $\rho(\mathcal{T}_L) = 2$, $\rho(\mathcal{T}_F) = 3$, and $\rho(\mathcal{T}_{LF}) = 4$ [Fig. 2(c)]. We detect an internal state when observing the follower, which is explained by observing both the leader and the follower, so that we infer a coupling from the leader to the follower.

Application of the method to Markovian synergistic dynamics—We consider the paradigmatic example of the synergistic system by James *et al.* [42],

$$X_{k+1} = \begin{cases} 0, & \text{w.p. } 1/2 \\ 1, & \text{w.p. } 1/2, \end{cases} \quad (10a)$$

$$Y_{k+1} = \begin{cases} 0, & \text{w.p. } 1/2 \\ 1, & \text{w.p. } 1/2, \end{cases} \quad (10b)$$

$$\text{and } Z_{k+1} = X_k \oplus Y_k, \quad (10c)$$

where $\oplus$ represents the XOR operator. In such a system, X and Y synergistically influence Z, that is, Z_{k+1} is dependent on $X_k(Y_k)$ only upon conditioning on $Y_k(X_k)$. This is a classical problem in stochastic dynamical systems and was used by James *et al.* [42] to illustrate how information theoretic measures can fail, as $\text{TE}_{X(Y) \rightarrow Z} = \text{MI}_{X(Y), Z} = 0$. Since pairwise interaction measures cannot be used to tackle such systems where the faithfulness assumption is violated, graphical methods such as PCMCi and its descendants also are not able to capture the structure [43]. Our method can infer the structure and conclude synergistic dynamics. Similar to the Boolean model, we measure the dynamics of units separately and in clusters (see Supplemental Material [30]). We obtain $\rho(\mathcal{T}_X) = \rho(\mathcal{T}_Y) = 2$, $\rho(\mathcal{T}_Z) = 3$, $\rho(\mathcal{T}_{XY}) = 4$, $\rho(\mathcal{T}_{XZ}) = \rho(\mathcal{T}_{YZ}) = 5$, and $\rho(\mathcal{T}_{XYZ}) = 8$, see Fig. 2(d). We detect an internal state when observing Z alone, which means that Z is influenced by other variables. We do not detect internal states while observing X, Y, or XY, which implies that no exterior variables are influencing either X or Y, or X and Y together. The internal state detected from Z cannot be explained by either X or Y separately. However, it can be explained by measuring them simultaneously, which implies that X and Y have a synergistic influence on Z.

Conclusion—In this Letter, we introduce a simple and intuitive method grounded in network science and control theory to detect directional coupling in a network dynamical system. Our approach successfully infers the structure of simple, yet challenging, models that often confound state-of-the-art techniques. For example, the robustness of our method to hidden dynamics addresses a fundamental challenge in classical causal inference, effectively tackling the well-known issue captured by the adage “Correlation does not imply causation” [44].

In addition to being robust to the presence of hidden units, the proposed method offers advantages in other challenging scenarios, such as contemporaneous coupling. In this case, the data sampling speed is slower than the interaction speed, therefore locking the system in a synchronous state. Methods based on the statistical dependence between time-shifted variables cannot correctly capture the direction of such interactions. While methods based on Pearl's causality might handle such cases, they have their own limitations, such as the faithfulness assumption, which our method partially overcomes, as demonstrated for the synergistic system [43]. While our method also comes with assumptions, we argue that they are fulfilled in many practical problems. For example, we assume that units' evolutions depend on both interactions and self-dynamics, which is standard in collective behavior [25,45].

The mathematical proof we developed to support the validity of our method strictly applies only to continuous-time, linear, and autonomous systems; yet, we have provided evidence that this tool can still work for nonautonomous, weakly nonlinear, and stochastic systems in discrete or continuous time. Straining the assumptions of the method for tackling other cases is not the only limitation of this study, whereby we identify three practical limitations. First, the type of data needed for building the detection matrix must be out of equilibrium, as we need to inspect the transient dynamics of the systems. Second, the detection matrix is sensitive to construction parameters, such as the sampling frequency [32]. Third, the proposed method cannot distinguish between direct and indirect influence. For example, the structures $Z \rightarrow X \rightarrow Y$ and $Y \leftarrow Z \rightarrow X \rightarrow Y$ are indistinguishable, meaning that we cannot assess if Z has a direct path to Y or it is just through X (see Supplemental Material [30]).

Future work should focus on many fronts. We should rigorously formulate the theory for the application of the method to more general dynamics and real-world data. Towards these aims, we should also extend the method to highly nonlinear systems [46], and possibly develop statistical tests to assess significance when working with experimental data. Finally, we should develop control-theoretic measures to quantify the extent of coupling, by relying on tools such as the observability Gramian [47] and observability radius [48].

Overall, this Letter lays the foundation for coupling detection and causal discovery from a control-theoretic perspective. To the best of our knowledge, it is the first to relate Kalman's observability to causal discovery, opening a new research avenue in complex systems.

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